CIE-407T — Distributed Systems & Cloud Computing

MID-SEM Study Pack: UNIT-I + UNIT-II ONLY

BPIT (IT, Sem VII) · GGSIPU (USICT handbook, batch 2021-22 onwards)

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0. How to use this pack

File	What is inside
`00_Syllabus_Checklist_and_Guide.md`	This file – checklist, source labels, exam pattern
`01_Unit1_Notes.md`	Complete Unit-I notes (Intro, models, IPC, RMI, RPC, events, Java RMI)
`02_Unit2_Notes.md`	Complete Unit-II notes (time, clocks, global state, mutual exclusion, election, multicast, consensus)
`03_Faculty_Emphasis_and_Quick_Tables.md`	Faculty-emphasis ranking, definitions, comparisons, diagrams, algorithms
`04_Question_Bank_with_Answers.md`	Question bank (questions first, model answers after)
`05_Predicted_Practice_and_Top30.md`	Extra "Claude Predicted Practice Questions" + Top-30 list
`06_Model_Papers.md`	3 model mid-sem papers + answer key
`07_Revision_Viva_and_Writing.md`	One-day revision sheet, 30 viva Qs, answer-writing format

Suggested order: Notes (files 01, 02) → Faculty emphasis (03) → attempt questions in 04 *without looking at answers* → 05 → model papers under time → 07 the night before.

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1. Source labels used everywhere

Label	Meaning
📘 **[Official syllabus]**	The topic is written in the GGSIPU syllabus PDF you uploaded
📝 **[Assignment-emphasized]**	The topic/question appears in your BPIT Assignment-1 or Assignment-2
➕ **[Additional / exam-oriented]**	NOT named in the syllabus, but added because textbooks (Coulouris – Distributed Systems: Concepts and Design; Ghosh; Tanenbaum) cover it under the same topic and examiners often expect it
🎯 **[Claude Predicted Practice Question]**	A question I created for practice. **These are predicted/practice questions based on syllabus and emphasis, not leaked or guaranteed questions.**

> ⚠️ Nothing here is guaranteed to appear. The notes are built to make you *understand* the topics so you can answer any variation.

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2. What the syllabus PDF actually says (verbatim summary)

Paper: CIE-407T, Distributed Systems and Cloud Computing, L-3, Credits 3. Theory: 25 marks internal (TCE) + 75 marks term-end.

UNIT-I – Introduction to Distributed Systems: Characteristics of distributed systems – Introduction; Examples of distributed systems (client-server, peer-to-peer, grid and cloud computing); Advantages of distributed systems; System models – Introduction, Architectural and Fundamental models; Networking and Internetworking; Interprocess Communication (message passing and shared memory); Distributed objects and Remote Method Invocation; RPC; Events and notifications; Case study – Java RMI.

UNIT-II – Synchronization: Time and Global States – Introduction; Logical clocks; Synchronizing physical clocks; events and process states; logical time and logical clocks; global states; distributed debugging. Coordination and Agreement: Distributed mutual exclusion; Elections; Multicast communication; consensus and related problems.

Textbooks (from syllabus): (1) Coulouris, Dollimore, Kindberg – *Distributed Systems: Concepts and Design*, 4th ed., Pearson; (2) Buyya – *Cloud Computing Principles and Paradigms*; (3) S. Ghosh – *Distributed Systems*, CRC Press 2010. **References:** Tanenbaum & van Steen; Kshemakalyani & Singhal; Gerard Tel.

Term-end paper pattern (from the handbook) – useful to know how questions are framed: - 9 questions in total. Q1 compulsory, objective/short, covers the entire syllabus, **15 marks**. - 4 units, each unit has 2 questions (attempt any 1), each question can have up to 5 sub-parts, **each unit = 15 marks**. - Standard of questions = level of the prescribed textbook.

> Note: your **mid-sem** is an internal BPIT exam, so the exact pattern may differ. Ask your faculty for marks/duration. This pack prepares 1-mark to 10-mark answers so that any pattern is covered.

Out of scope for mid-sem – "Outside Mid-Sem Syllabus": Unit-III (Distributed File Systems, Hadoop, load balancing/sharing, task assignment) and Unit-IV (Cloud Computing layers, IaaS/PaaS/SaaS, migration). Only the *idea* of "cloud computing" as an **example of a distributed system** is in Unit-I, so it is covered briefly there.

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3. MID-SEM SYLLABUS CHECKLIST (tick as you finish)

UNIT-I

#	Topic (as in your brief)	Where	Assignment link	Done
1	Introduction to Distributed Systems	01 §1	A1-1a	☐
2	Characteristics of Distributed Systems	01 §2	A1-1a	☐
3a	Example: Client–Server	01 §3.1	A1-1b	☐
3b	Example: Peer-to-Peer	01 §3.2	A1-1b	☐
3c	Example: Grid Computing	01 §3.3	A1-1b	☐
3d	Example: Cloud Computing (as an example only)	01 §3.4	A1-1b	☐
3e	Comparison CS vs P2P vs Grid vs Cloud	01 §3.5	A1-1b	☐
4	Advantages of Distributed Systems	01 §4	–	☐
5a	System Models – Introduction	01 §5.1	A1-2a	☐
5b	Architectural Models	01 §5.2	A1-2a	☐
5c	Fundamental Models (interaction, failure, security)	01 §5.3	A1-2a	☐
6	Networking and Internetworking	01 §6	A1-2b	☐
7a	IPC – Message Passing	01 §7.1	A1-3a	☐
7b	IPC – Shared Memory	01 §7.2	A1-3a	☐
7c	Message Passing vs Shared Memory	01 §7.3	A1-3a	☐
8	Distributed Objects	01 §8	A1-3b	☐
9	Remote Method Invocation (RMI)	01 §9	A1-3b, A1-1c	☐
10	RPC	01 §10	(indirect)	☐
10b	RMI vs RPC	01 §10.5	–	☐
11	Events and Notifications	01 §11	–	☐
12	Case Study: Java RMI	01 §12	A1-1c	☐

UNIT-II

#	Topic	Where	Assignment link	Done
1	Introduction to Synchronization	02 §1	A2-1a	☐
2	Time and Global States (intro)	02 §2	A2-1a	☐
3	Logical Clocks	02 §5	A2-1b	☐
4	Synchronizing Physical Clocks (Cristian, Berkeley, NTP)	02 §7	A2-1c	☐
5	Events and Process States	02 §3	A2-1a	☐
6	Logical Time and Logical Clocks (Lamport, Vector)	02 §4–6	A2-1b	☐
6b	Logical vs Physical time	02 §8	–	☐
7	Global States (cut, snapshot)	02 §9	A2-2a, 2b	☐

#	Topic	Where	Assignment link	Done
8	Distributed Debugging	02 §10	–	☐
9	Coordination and Agreement (intro)	02 §11	–	☐
10	Distributed Mutual Exclusion	02 §12	A2-3a	☐
11	Election Algorithms (Bully, Ring)	02 §13	A2-3b	☐
12	Multicast Communication	02 §14	–	☐
13	Consensus and Related Problems	02 §15	A2-4a, 4b	☐

Coverage statement: every one of the 21 Unit-I and 13 Unit-II items in your brief has a section in the notes.

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4. Symbols used in diagrams

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P1, P2, P3   = processes            ──►  message / arrow
[ ]          = a box (component)    - ►   reply / return
●            = event on timeline    ✗    crashed process
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Hinglish tips are marked with 🗣️.